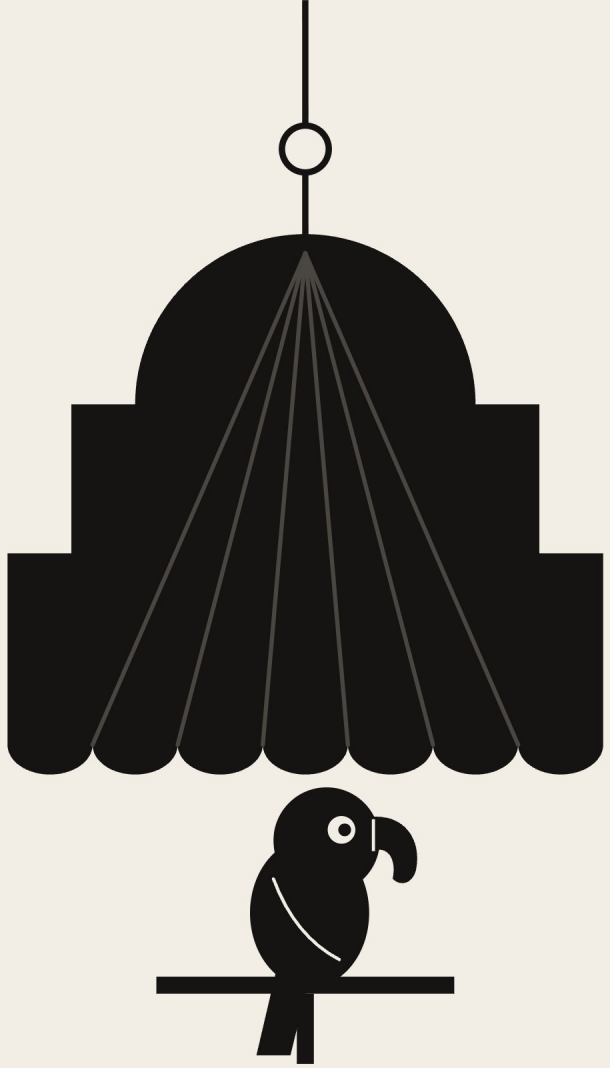
Evaluating AI Agents

Workshop by Vladislav Kutsevalov and Lidia Barkanova

14 October · Bilbao



Preparation

1 · Get a free Google Gemini API key

Go to aistudio.google.com
Click “API keys” in the left menu
→ “Create API key”

2 · Make sure you have the basics

Git git-scm.com/downloads
Python 3.12+ python.org/downloads
A code editor e.g. VS Code

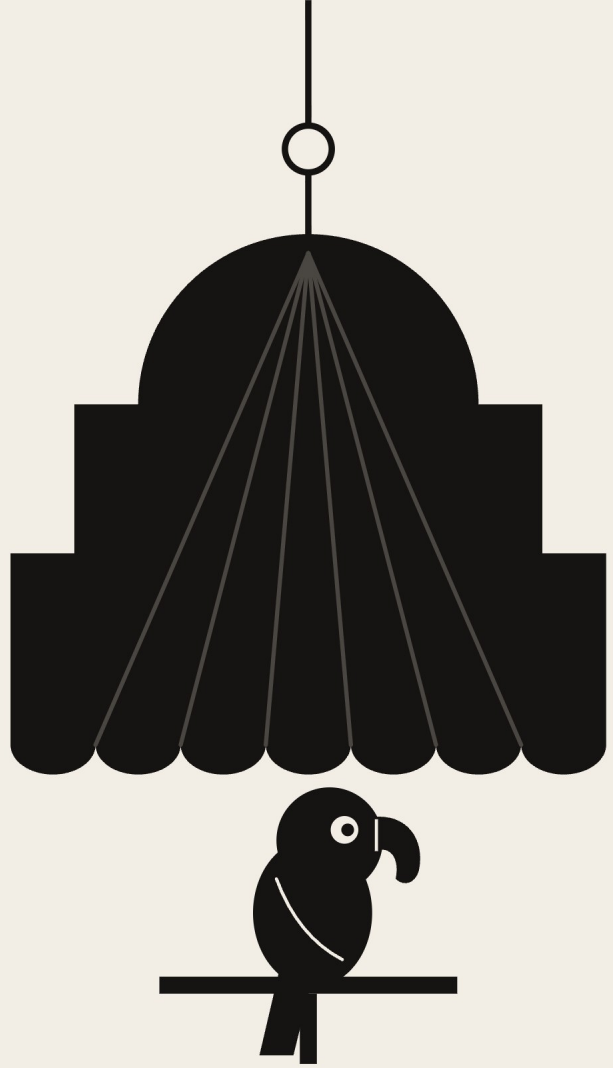
Rules

● **Red sticker** — I need help ○ **Green sticker** — all good **No AI for testing**

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Python 3.12+ python.org/downloads
A code editor e.g. VS Code

3 · Clone the repo

github.com/teamlida/eval-ai-agents

4 · Install Google ADK

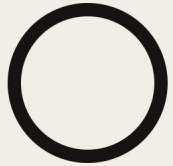
pip install google-adk

Rules



Red sticker

I need help — one of us comes to you



Green sticker

All good — done, ready to move on

No AI for testing



Vladislav Kutsevalov

Staff Quality Engineer · Infobip



Lidia Barkanova

Senior Software Engineer · Super

AGENTIC AI

01

Rule-based

Follows scripted if–then rules. Only answers what was pre-programmed.

02

Conversational

"Understands" what you mean and keeps a dialogue, but picks from prepared answers.

03

Generative

An LLM writes new text on the fly. It talks, but doesn't act.

04

Agentic

An LLM that calls tools in a loop to get a task done.

platform.openai.com/tokenizer

Generations after idsalliance.org

AI agent =

Tools + Loop

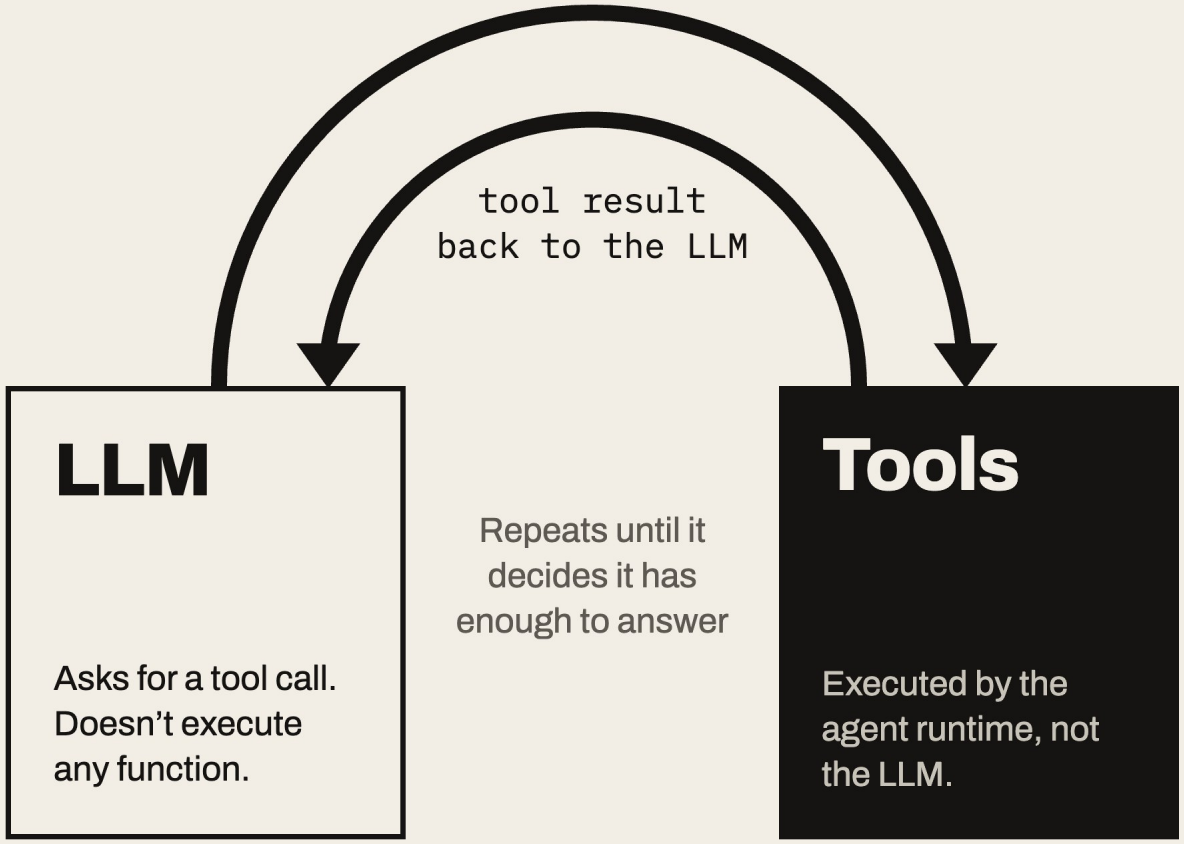
+ secondary parts

Instructions
the system prompt: the goal and the rules

Context
what it knows in this conversation

State
the history of the session

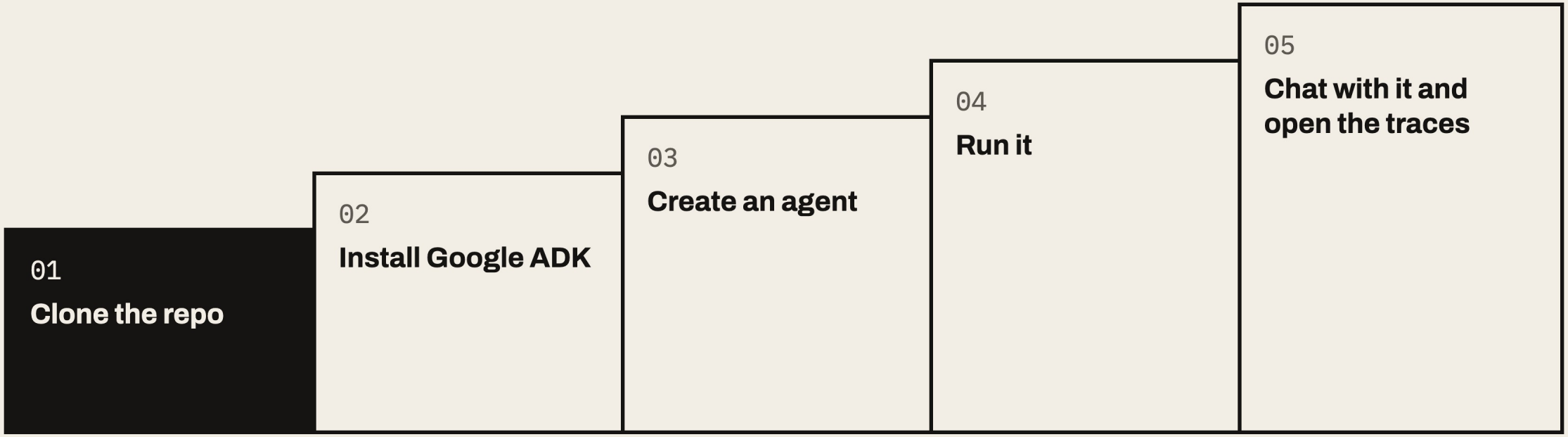
Restrictions
ways to restrict agent actions/response



A simple AI agent has one tool.
You ask a question that needs this tool.

How many times does it call the LLM?

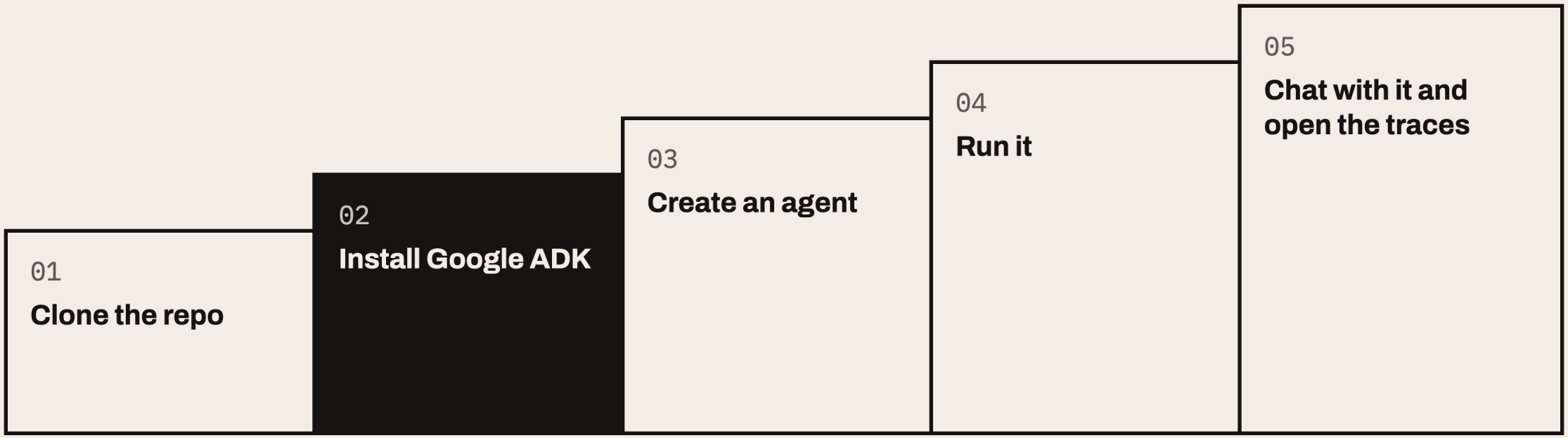
Let’s look closer



https://github.com/teamlida/eval-ai-agents

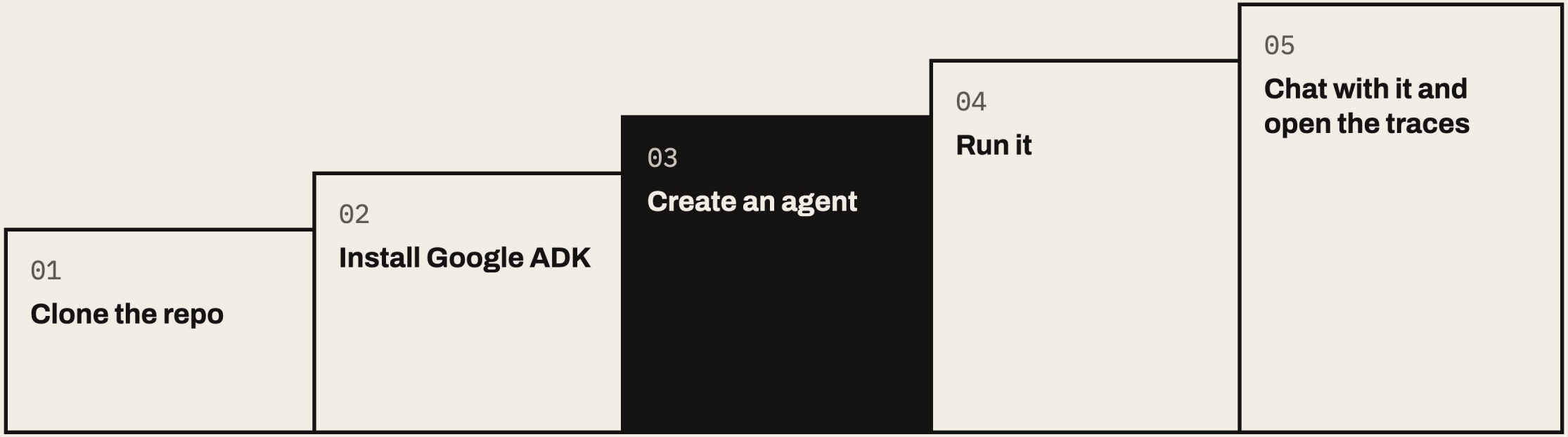
step 01

Let’s look closer



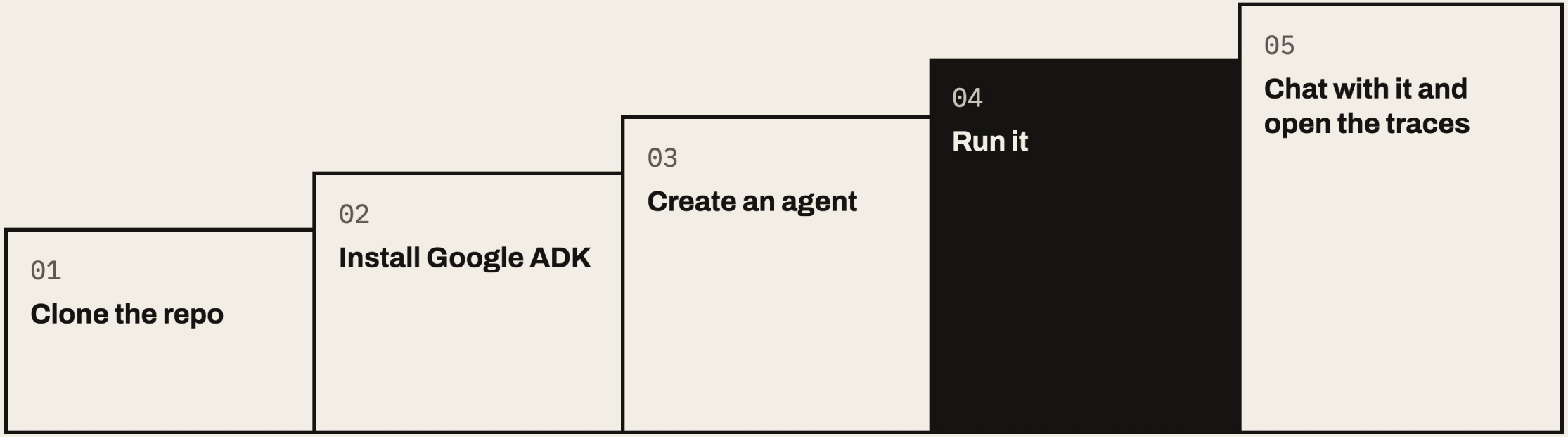

```
pip install google-adk                                    step 02
```

Let’s look closer



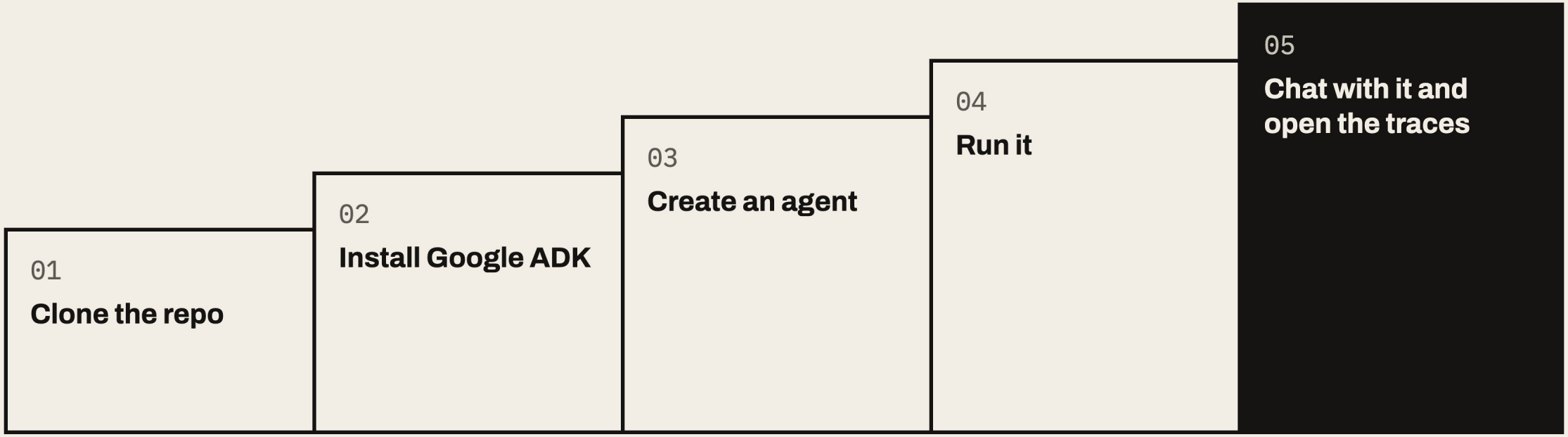

```
adk create my_agent                                    step 03
```

Let’s look closer



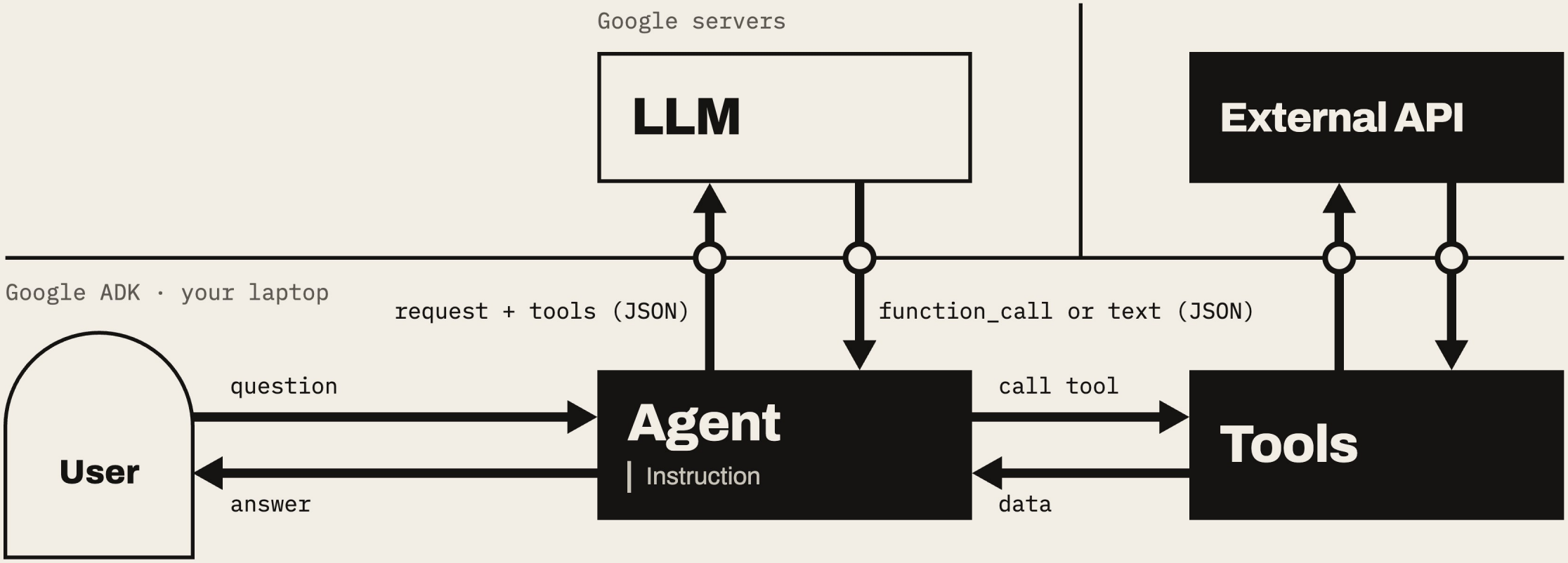

```
adk web --port 8000                                    step 04
```

Let’s look closer



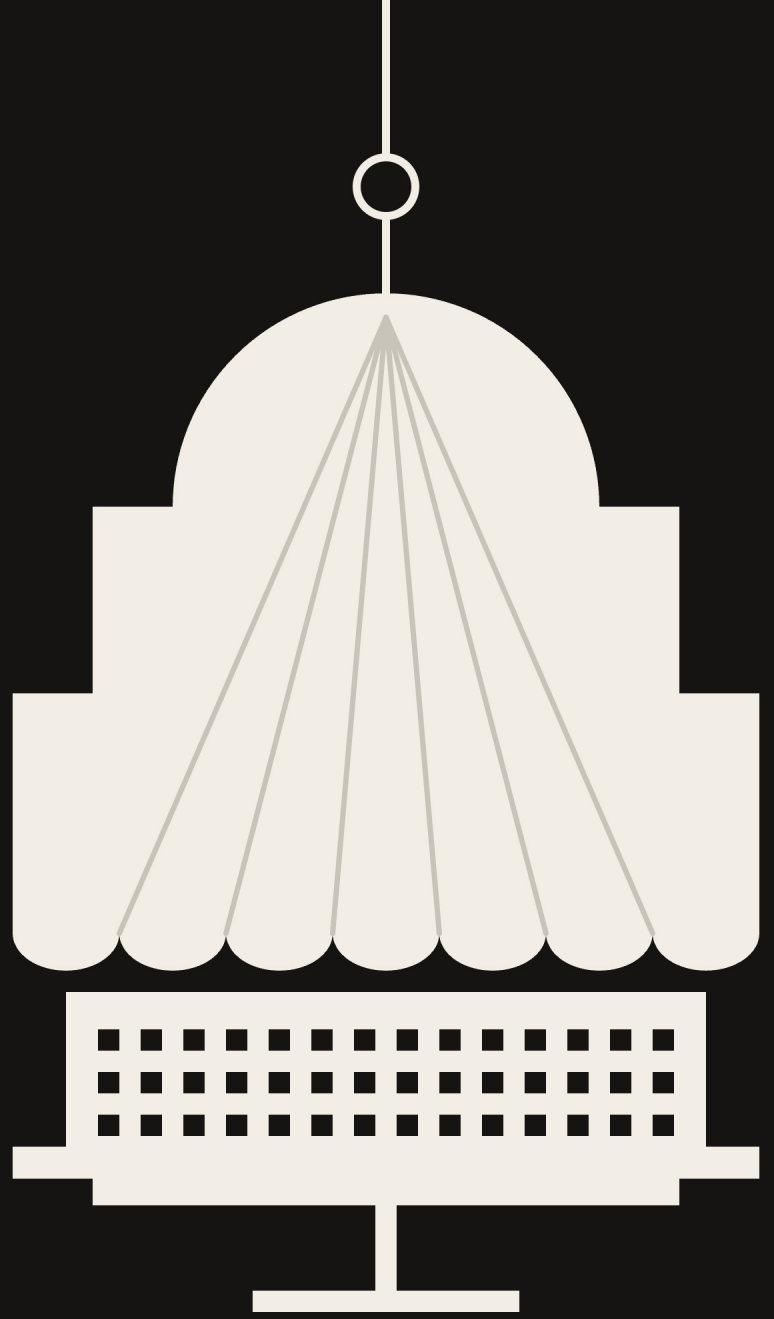

```
localhost:8000 → chat → open the traces                    step 05
```

How it works



The LLM never touches the API. ADK does. The LLM only writes JSON.

Parrot
in the Dark



Break

30 minutes

```
pip install "google-adk[eval]"
```



What could go wrong?

10-min exercise

01 **Take the sticker**

02 **Write an example of a problem or a risk with AI agent**

03 **Put it on a board**



Product Quality criteria

01 **Capability** Can it perform the required functions?	02 **Reliability** Will it work well and resist failure?	03 **Usability** How easy is it for real users?	04 **Charisma** How appealing is the product?	05 **Security** How well is the product protected against unauthorized use?
06 **Scalability** How well does it scale?	07 **Compatibility** How well does it work with external parts?	08 **Performance** How speedy and responsive is it?	09 **Installability / Activation** How easily can it be enabled?	10 **Development** How well can we build, test, and modify it?

James Bach · Heuristic Test Strategy Model

Selected LLM syndromes

Not doing enough

Doesn’t ask questions
Doesn’t answer the question
Says nothing useful
Forgets what it said

Doing the wrong thing

Wrong answers
Makes things up
Says what it shouldn’t
Offensive answers

Unreliable

Gives in when you push back
Different answer every time
Breaks its own rules

Hard to fix

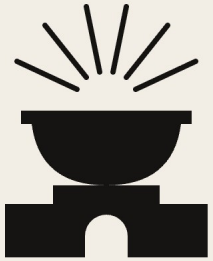
Doesn’t learn from fixes
Fixes break other things
Chases goals nobody asked for

Michael Bolton & James Bach, “What Are We Thinking in the Age of AI?”, 2025
developsense.com/presentations/2025-10-WhatAreWeThinkingInTheAgeOfAI-Ljubljana.pdf



valere

Latin — “to be strong, to be worth”



testum

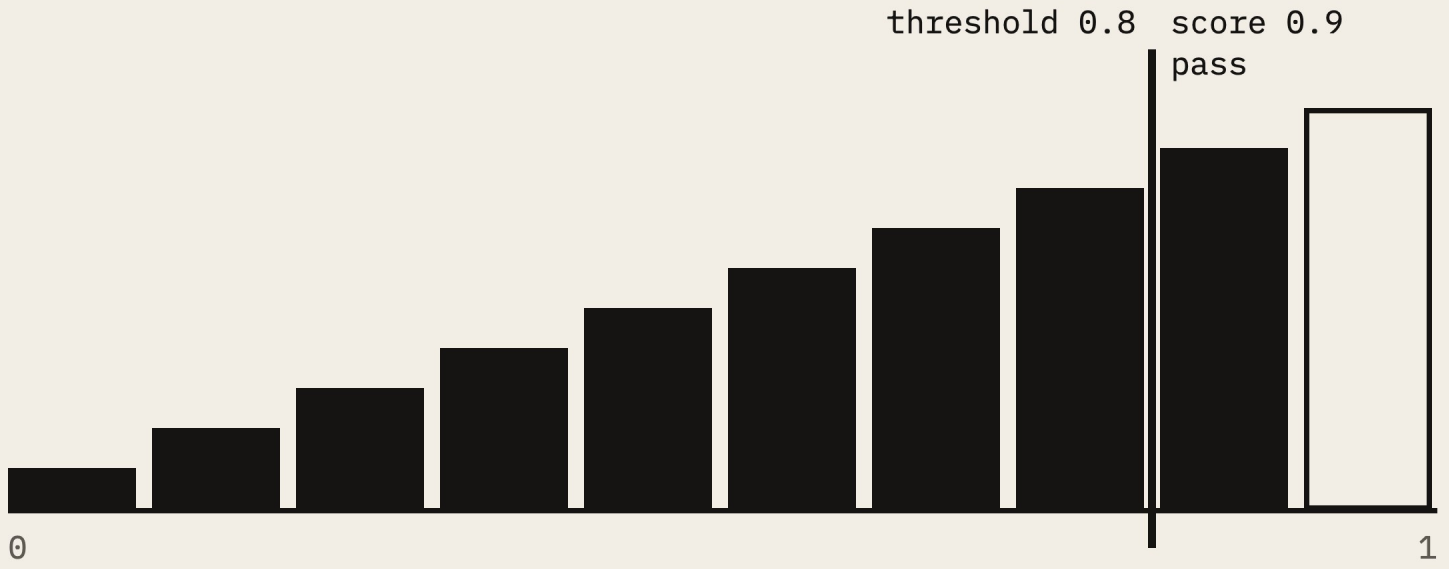
Latin — “an earthen pot”

Testing vs evaluation?

Evaluation of AI Agent - repeatable measurement against criteria with a score and a threshold.

criteria

The response contains the user's personal data.



Two kinds of checks

code checks it

Deterministic

The tool was called with the correct parameters

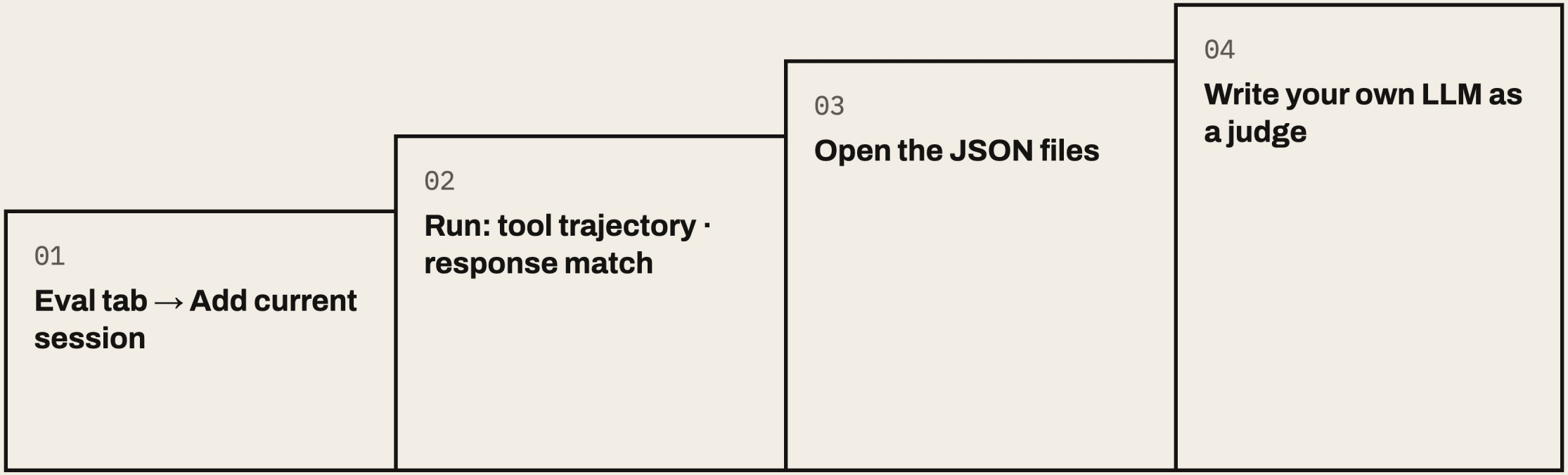
a "judge" checks it

Non-deterministic

The LLM answer fits a criterion

Evals in ADK

```
pip install "google-adk[eval]"
```

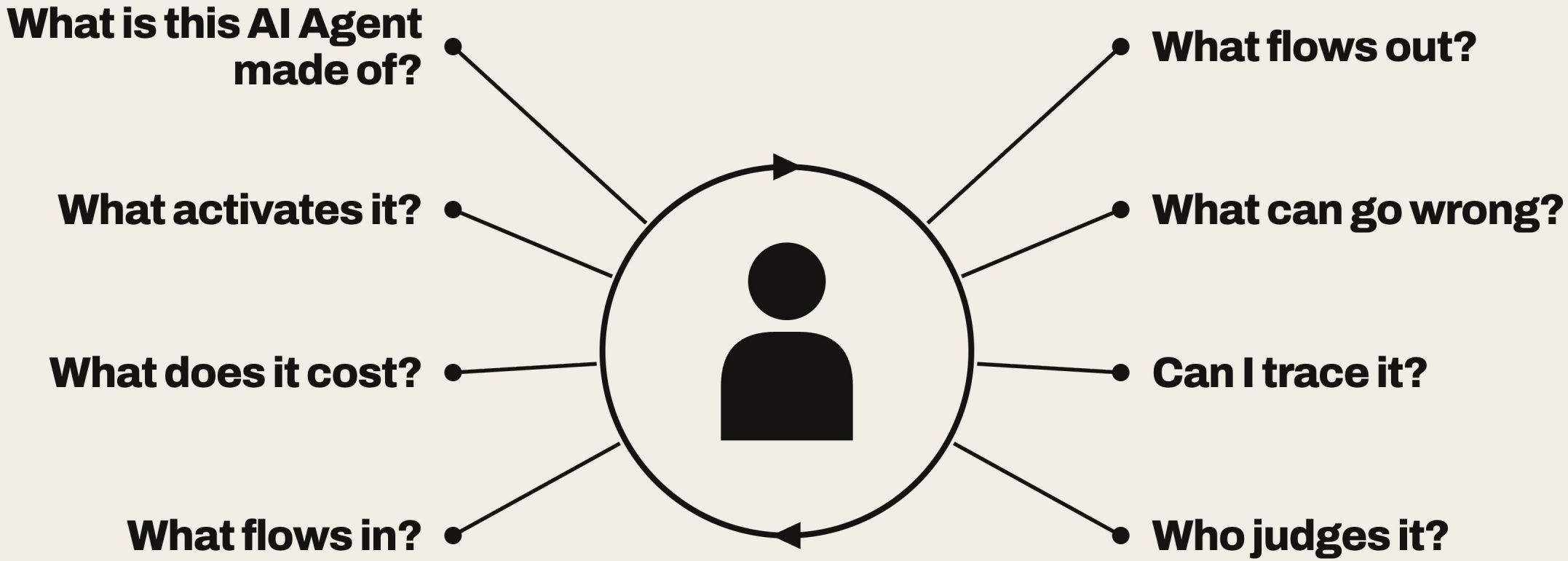


PAIR TESTING Bug hunt: Star Wars Guide

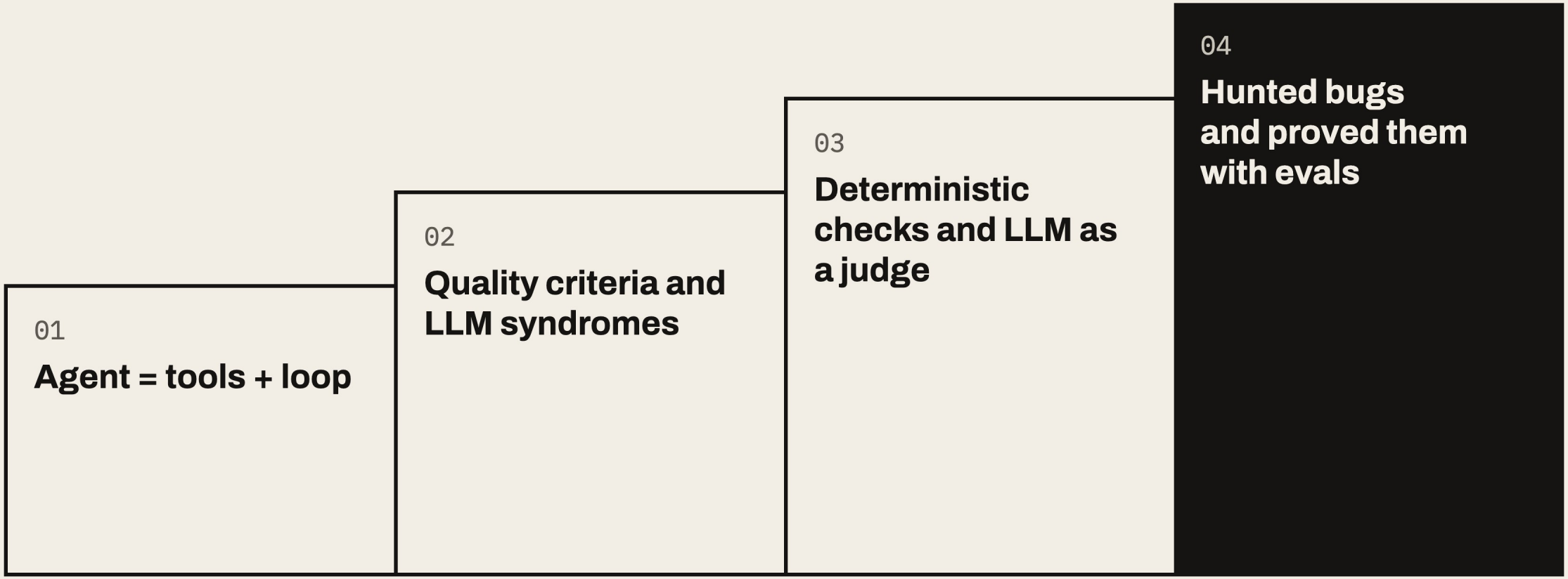
- Working in pairs: one person execute another register findings and guides
- Don’t open agent.py
- Talk to the agent in ADK, read the traces
- Every suspected bug → an eval case (UI or JSON)

Out of API quota? Create new project in Google AI Studio and get a new key · Stuck? Red sticker

Human in the loop



What we did today



Thank you!

https://github.com/teamlida/eval-ai-agents/tree/main/materials

Vladislav · LinkedIn

Lidia · LinkedIn

Feedback

